

ENGINEERING_RULES.md

This repository follows professional engineering standards.

Code Organization

Prefer:

services/
repositories/
controllers/
domain/

Avoid large files with mixed responsibilities.

Naming

Names must be:

- descriptive
- consistent
- domain oriented

Avoid abbreviations.

Error Handling

Never return raw stack traces.

Errors must be:

- categorized
- logged
- safe for clients

Logging

Logs must include:

- timestamp
- context
- operation
- result
- error details when applicable

Refactoring

When refactoring:

- maintain behavior compatibility
- ensure tests pass
- reduce complexity

Performance

Optimize only when:

- measurable bottlenecks exist
- complexity analysis supports change

Dependency Management

New dependencies must:

- solve a clear problem
- be widely maintained
- not duplicate existing functionality